

7. $\mathbb{C}$, the set of all complex numbers.

Intervals: Sets of all the real numbers between two numbers a and b , with or without a and b .

$$[a, b] = \{x \mid a \leq x \leq b\} \rightarrow \text{Closed interval}$$

$$[a, b) = \{x \mid a \leq x < b\}$$

$$(a, b] = \{x \mid a < x \leq b\}$$

$$(a, b) = \{x \mid a < x < b\} \rightarrow \text{Open interval}$$

Example 5: The set $\{N, Z, Q, R\}$ is a set containing four elements, each of which is a set. The four elements of this set are N , the set of natural numbers; Z , the set of integers; Q , the set of rational numbers; and R , the set of real numbers. $\square$

Very Important Concept

Remark: Note that the concept of a datatype, or type, in computer science is built upon the concept of a set. In particular, a datatype or type is the name of a set, together with a set of operations that can be performed on objects from that set.

• For example, boolean is the name of the set $\{0, 1\}$, together with operators on one or more elements of this set, such as AND, OR and NOT.

Definition 2: Two sets are equal if and only if they have the same elements. Therefore, if A and B are sets, then A and B are equal if and only if $\forall x (x \in A \leftrightarrow x \in B)$. We write $A = B$ if A and B are equal sets.

Example 6: The sets $\{1, 3, 5\}$ and $\{3, 5, 1\}$ are equal, because they have the same elements. Note that the order in which the elements of a set are listed does not matter. Note also that it does not matter if the elements of a set are listed more than once, so $\{1, 3, 3, 3, 5, 5, 5, 5\}$ is the same as the set $\{1, 3, 5\}$ because they have the same elements. $\square$